Krishna Lal

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SUMMARY

AI/ML engineer (B.Tech CSE Data Science, CGPA 8.7) focused on computer vision and deep learning: built a YOLOv8 + OpenCV detection, tracking and event pipeline (12 tracked objects, 8.04 FPS on a laptop) and trained a 4.8M-parameter transformer from scratch in PyTorch. Strong in Python, NumPy, Pandas, Git and REST APIs (FastAPI, Flask), with 1 production e-commerce platform shipped in 1 month. Open-source contributor (ISTE Open Source 101 Project Admin, Resourcio Aperture 3.0, ELUSOC), leader of a 2000+ member tech community. 1st place InnovateX 2025, 3rd place National Ideathon 2025, Top 10 MLH Diversion 2026.

EXPERIENCE

Full Stack AI & ML Intern – Tivsa BlacksandWhite Pvt. Ltd. (blacksandwhite.in) | Oct 2026 – Nov 2026 (1 month) | Remote

- Built the company's entire multi-vendor e-commerce website from scratch in 1 month, with 15+ features live: product catalogue, cart, checkout, customer accounts with profile editing and saved addresses, seller login and dashboard, and an About Us page.
- Developed a single-page monochrome neumorphic storefront (HTML, CSS, JS) on the WooCommerce Store API plus a custom BSW REST plugin (PHP, Python backend).
- Deployed an LLM-based RAG chatbot for 24/7 customer support and applied ML models to improve customer retention rate (CRR).
- Red-teamed the RAG chatbot with 25+ prompt-injection attempts (instruction override, system-prompt leakage, data exfiltration, jailbreaks), found 5 weaknesses and fixed them with input filtering and system-prompt hardening.
- Managed the brand's social media presence alongside the engineering work.

Vice President – GRID Community | 2025 – Present

- Lead operations of a 2000+ member technical community.
- Organised 6 industry-speaker events end-to-end and co-organised the Hacktoberfest 26 Hack Day (30 Oct 2026) with MLH.
- Drove membership growth through engagement and outreach programmes.

EDUCATION

Haldia Institute of Technology – B.Tech, CSE (Data Science) | 2024 – 2028 | CGPA 8.7 (up to 5th semester)

PROJECTS

Real-Time CCTV Object Detection & Event Pipeline (GitHub)

- Built a YOLOv8n + OpenCV pipeline for detection, multi-object tracking and polygon restricted-zone entry detection, with live FPS monitoring.
- Converted 1,832 per-frame detections into 2 structured JSON events across 12 unique tracked objects (300 frames, 8.04 FPS end-to-end on an HP 15s laptop), using warm-up and cooldown logic to suppress duplicate events.
- Tools: Python, OpenCV, Ultralytics YOLOv8, NumPy, JSON

Vector – From-Scratch GPT-Style Language Model & Voice AI (GitHub)

- Trained a 4.8M-parameter decoder-only transformer (6 layers, 8 heads) from scratch in PyTorch with a custom tokenizer, reaching validation perplexity of about 2.19.
- Added a deterministic arithmetic tool that lifts accuracy from 13.6% to 100%, plus a speech-to-text-to-speech voice interface.
- Tools: Python, PyTorch, Custom Tokenizer, SpeechRecognition, pyttsx3

SKILLS

Computer Vision: OpenCV, YOLOv8 (Ultralytics), object detection, multi-object tracking, video processing pipelines, event detection
ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Transformers, Feature Engineering, Model Training & Evaluation
Generative AI: LLMs, RAG, LangChain, LangGraph, AI Agents, prompt-injection testing
Languages: Python, C++, Java, SQL, Bash, JavaScript, PHP, HTML/CSS
Engineering: REST APIs, FastAPI, Flask, Docker, MLflow, CI/CD, AWS, Linux, Git/GitHub